



Turning in Code

Ontario Math C3.1 — repeating events (move + turn)

Name:

Date:

★ I can read a loop that makes Bit move and turn.

Move, then turn

Last time Bit moved in a straight line. Now we add a turn inside the loop, so Bit moves AND turns each time around.

The turn amount is given to you (a square corner is right 90). Read the loop and see the path Bit draws.

The block you know

repeat 3

move 50

turn right 90

Move 50, turn right 90 — three times.

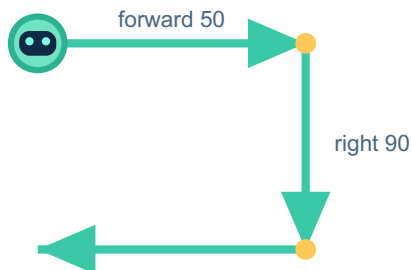
The same thing in Python

PYTHON

```
for step in range(3):  
    bit.forward(50)  
    bit.right(90)
```

OUTPUT

Bit draws 3 sides with 3 right-angle turns.



range(3) draws 3 sides and 3 turns — one more side would close a square.

1 Read it

In the loop, how many times does Bit move forward? _____ How many times does Bit turn? _____

2 Predict, then run

You change `range(3)` to `range(4)` (forward 50, right 90 stay the same). PREDICT what shape Bit draws now. Then run it to check.

3 Write your own

Make Bit move and turn right 90, FOUR times. Fill in the numbers: for step in `range(_____)`:
`bit.forward(50) bit.right(_____)`

4 Challenge

In for step in `range(4)`: `bit.forward(50); bit.right(90)` — Bit turns 90 each time. How many degrees does Bit turn in TOTAL? (Hint: it goes all the way around.)

Big idea

Putting a move AND a turn inside the loop lets Bit draw shapes. The turn amount decides the corner; the number of repeats decides how many sides.



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Quick facilitation notes & answer key

What this worksheet teaches

Now a turn is added inside the repeat, so Bit moves AND turns each time around. Students see that repeating "move, then turn" draws the sides of a shape. The turn amounts are given. No coding experience needed.

Before you start — showing the drawing (optional)

The worksheet shows the typed instructions AND a picture of the sides Bit draws, so you can teach straight from the printout — no computer required.

To show it live (optional): open a browser, go to a free "Python turtle" playground (e.g. search "trinket python turtle"), type the few lines from the worksheet, and press Run — a turtle draws it. You only press Run.

How to lead it (about 30 minutes)

1. Recall last sheet: a repeat that only moved drew a straight line. Now we add a turn.
2. Show it (you do). Read it aloud: "repeat 3 times: move forward 50, then turn right." Bit draws 3 sides with 3 turns.
3. Let them work. Students do Exercises 1–4, solo or in pairs.
4. Wrap up: putting a move AND a turn inside the repeat lets Bit draw shapes; the repeat number sets how many sides.

Answer key (these are the verified correct answers)

Exercise 1 — 3 forwards and 3 turns.

Exercise 2 — a square (repeating 4 times adds the 4th side and closes it).

Exercise 3 — the missing numbers are 4 (repeat 4 times) and 90 (turn right 90).

Exercise 4 (challenge) — 360° in total ($4 \times 90 =$ one full turn).

If students get stuck — and ways to adjust

Watch for: thinking more repeats means a bigger shape — more repeats means more sides, not a bigger size.

Make it easier: trace the 3 sides on the picture before counting.

Make it harder: ask what shape "repeat 4: move forward 50, turn right 90" makes (a square).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The student counts the forwards and turns (Ex 1), predicts the square when the repeat changes to 4 (Ex 2), fills in the numbers (Ex 3), and finds the total turn (Ex 4).